Analyzing the k Most Probable Solutions in EDAs Based on Bayesian Networks

Carlos Echegoyen, Alexander Mendiburu, Roberto Santana, and Jose A. Lozano

Abstract. Estimation of distribution algorithms (EDAs) have been successfully applied to a wide variety of problems but, for the most complex approaches, there is no clear understanding of the way these algorithms complete the search. For that reason, in this work we exploit the probabilistic models that EDAs based on Bayesian networks are able to learn in order to provide new information about their behavior. Particularly, we analyze the k solutions with the highest probability in the distributions estimated during the search. In order to study the relationship between the probabilistic model and the fitness function, we focus on calculating, for the k most probable solutions (MPSs), the probability values, the function values and the correlation between both sets of values at each step of the algorithm. Furthermore, the objective functions of the k MPSs are contrasted with the k best individuals in the population. We complete the analysis by calculating the position of the optimum in the k MPSs during the search and the genotypic diversity of these solutions. We carry out the analysis by optimizing functions of different natures such as Trap5, two variants of Ising spin glass and Max-SAT. The results not only show information about the relationship between the probabilistic model and the fitness function, but also allow us to observe characteristics of the search space, the quality of the setup of the parameters and even distinguish between successful and unsuccessful runs.

Carlos Echegoyen · Alexander Mendiburu · Jose A. Lozano
Intelligent Systems Group, Department of Computer Science and Artificial Intelligence, University of the Basque Country, Paseo Manuel de Lardizábal 1, 20080 San Sebastián - Donostia, Spain
e-mail: {carlos.echegoyen,alexander.mendiburu,ja.lozano}@ehu.es

Roberto Santana
Universidad Politécnica de Madrid, Campus de Montegacedo sn. 28660. Boadilla del Monte, Madrid, Spain
e-mail: roberto.santana@upm.es

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1 Introduction

Estimation of distribution algorithms (EDAs) [23, 32] are an evolutionary computation branch which have been successfully applied in problems of different domains [2, 27, 40]. However, despite their successful results there are a wide variety of open questions [43] regarding the behavior of this type of algorithms. Therefore, it is necessary to continue studying EDAs in order to better understand how they solve problems and advance their development.

The main characteristic of EDAs is the use of probabilistic models instead of the typical crossover and mutation operators employed by genetic algorithms [17]. Thus, linkage learning, understood as the ability to capture the relationships between the variables of the optimization problem, is accomplished in EDAs by detecting and representing probabilistic dependencies using probability models. This type of algorithms uses machine learning methods to extract relevant features of the search space through the selected individuals of the population. The collected information is represented using a probabilistic model which is later employed to generate new solutions. In this way, a learning and sampling iterative process is used to lead the search to promising areas of the search space.

The models employed to encode the probability distributions are a key point in the performance of EDAs. In this regard, probabilistic graphical models [7] are a powerful tool, and in particular, Bayesian networks have been extensively applied in this field [15, 29, 38]. One of the benefits of EDAs that use these kind of models is that the complexity of the learned structure depends on the characteristics of the selected individuals. Additionally, the Bayesian networks learned during the search are suitable for human interpretation, helping to discover unknown information about the problem structure. In fact, a straightforward form of analyzing EDAs based on Bayesian networks is through the explicit interactions among the variables they provide. In this sense, it has been shown how different parameters of the algorithm influence the structural model accuracy [25], how the dependencies of the probabilistic models change during the search [19] and how the networks learned can provide information about the problem structure [11, 14, 19].

In previous works [12, 13], we took a different path in the study of EDAs by recording probabilities and function values of distinguished solutions of the search space during the run. In this chapter, we extend that work focusing on the k most probable solutions in the probability distribution encoded by the probabilistic models at each generation of an estimation of Bayesian network algorithm (EBNA) [15]. By collecting different measurements, we analyze these solutions using four classes of test problems which have been widely used in EDAs: Trap5, two-dimensional Gaussian Ising spin glasses, two-dimensional $\pm J$ Ising spin glasses and Max-SAT. We argue that the analysis proposed supports a different perspective that is able to reveal patterns of behavior inside this type of algorithms. Furthermore, we show that the probabilistic models, besides structural information about the problem, can contain useful information about the function landscape and also about the quality of the search. Thus, the results provided in this work promote the exploitation of linkage learning in order to learn more about the algorithms and advance their development.

The rest of the chapter is organized as follows. Section 2 introduces Bayesian networks, the calculation of the k most probable solutions and presents estimation of Bayesian network algorithms. Section 3 explains the experimental design. Sections 4, 5, 6 and 7 discuss Trap5, Gaussian Ising, $\pm J$ Ising and Max-SAT problems respectively, analyzing the behavior of the k most probable solutions during the optimization process of each function. Section 8 discusses relevant previous works and finally, Section 9 draws the conclusions obtained during the study.

2 Background

2.1 Bayesian Networks

Formally, a Bayesian network [7] is a pair (S, θ) representing a graphical factorization of a probability distribution. The structure S is a directed acyclic graph which reflects the set of conditional (in)dependencies among n random variables $\mathbf{X} = (X_1, \ldots, X_n)$. On the other hand, θ is a set of parameters for the local probability distributions associated with each variable.

The factorization of the probability distribution is codified by S:

$$p(\mathbf{x}) = \prod_{i=1}^{n} p(x_i|\mathbf{pa}_i) \tag{1}$$

where $\mathbf{pa}_i$ denotes a value of the variables $\mathbf{Pa}_i$, the parent set of X_i in the graph S.

With reference to the set of parameters θ, if the variable X_i has r_i possible values, the local distribution $p(x_i|\mathbf{pa}_i^j, \theta_i)$ is an unrestricted discrete distribution:

$$p(x_i^k|\mathbf{pa}_i^j, \theta_i) \equiv \theta_{ijk} \tag{2}$$

where $\mathbf{pa}_i^1, \ldots, \mathbf{pa}_i^{q_i}$ denote the q_i possible values of the parent set $\mathbf{Pa}_i$. In other words, the parameter θ_{ijk} represents the probability of variable X_i being in its k-th value, knowing that the set of its parents' variables is in its j-th value. Therefore, the local parameters are given by $\theta_i = ((\theta_{ijk})_{k=1}^{r_i})_{j=1}^{q_i}$.

2.2 Learning Bayesian Networks from Data

In order to complete the Bayesian network learning, it is necessary to obtain a structure S and a set of parameters θ from a data set. Firstly, although there are different strategies to learn the structure, we focus on a method called "score + search" which is the one used in the experiments presented in this paper. This approach is based on a procedure to search high quality structures according to a determined metric or score. This score tries to measure how well a given structure S represents the underlying probability distribution of a dataset D. Particularly, in this work we use the Bayesian Information Criterion score (*BIC*) [44] based on penalized maximum likelihood. In order to make the search in a space of structures, we use Algorithm B [6] because it is able to return good results efficiently. Algorithm B is a greedy search procedure which starts with an arcless structure and, at each step, adds the arc with the maximum improvement in the score. The algorithm finishes when there is no arc whose addition improves the score.

Once the structure has been learned, the parameters of the Bayesian network are calculated using the Laplace correction:

$$\hat{\theta}_{ijk} = \frac{N_{ijk} + 1}{N_{ij} + r_i} \tag{3}$$

where N_{ijk} denotes the number of cases in D in which the variable X_i has the value x_i^k and $\mathbf{Pa}_i$ has its j^{th} value, and $N_{ij} = \sum_{k=1}^{r_i} N_{ijk}$.

2.3 Estimation of Distribution Algorithms Based on Bayesian Networks

Following the main scheme of EDAs, EBNA [15] works with populations of N individuals that constitute sets of N candidate solutions. The initial population is generated according to a uniform distribution, and hence, all the solutions have the same probability to be sampled. Each iteration starts by selecting a subset of promising individuals from the population. Although there are different selection methods, in this case we use truncation selection with threshold 50%. Thus, the $N/2$ individuals with the best fitness value are selected. The next step is to learn a Bayesian network from the subset of selected individuals. Once the Bayesian network is built, the new

Algorithm 1. EBNA

$BN_0 \leftarrow (S_0, \theta^0)$ where S_0 is an arc-less structure, and θ^0 is uniform
$D_0 \leftarrow$ Sample N individuals from BN_0
t $\leftarrow$ 1
do {
  $D_{t-1} \leftarrow$ Evaluate individuals
  $D_{t-1}^{Se} \leftarrow$ Select $N/2$ individuals from D_{t-1}
  $S_t^* \leftarrow$ Obtain a network structure
  $\theta^t \leftarrow$ Calculate θ_{ijk}^t using D_{t-1}^{Se} as the data set
  $BN_t \leftarrow (S_t^*, \theta^t)$
  $D_t \leftarrow$ Sample $N-1$ individuals from BN_t and create the new population
} **until** Stop criterion is met

population can be generated. At this point there are different possibilities. We use an elitist criterion. From the Bayesian network, $N-1$ new solutions are sampled and then mixed with the N individuals of the current population. The N best individuals, among the $2N-1$ available, constitute the new population. The procedure of selection, learning and sampling is repeated until a stop condition is fulfilled. A pseudocode of EBNA is shown in Algorithm 1.

2.4 Abductive Inference and Most Probable Configurations

Given a probabilistic graphical model, the problem of finding the most probable configuration (MPC), consists of finding a complete assignment with maximum probability that is consistent with the evidence. Similarly, the problem of finding the k most probable configurations consists of finding the k configurations with maximum probability in the distribution encoded by the graphical model. The name given to the most probable configurations changes between the different domains of application and according to the type of graphical models used. It is also known as the most probable explanation problem, and the n-best list [35] or n-best hypothesis problem [1]. In Bayesian networks, the process of generating the MPC is usually called abductive inference. In our context of EDAs, these points will be called the most probable solutions (MPSs).

Computing the MPC can be easily done on a junction tree by a simple message-passing algorithm [9]. However, the computation of the k MPCs requires the application of more complex schemes [34, 45]. In any case, when the inference is done by using junction trees the results are exact.

In this work we use the algorithm presented in [34]. It has been conceived for finding the most probable configurations in a junction tree and therefore gives us exact solutions. It is based on the use of a simple message-passing scheme and a dynamic programming procedure to partition the space of solutions. The algorithm starts by finding the most probable configuration, and from this configuration, the space of solutions is partitioned in disjunct sets for which the respective MPC is found. The second MPC is the one with the highest probabilities among all of the partitions. More details about the algorithm can be found in [34].

The algorithm used in this work has been implemented using the Bayes Net Toolbox [33] and can be found in [42].

3 Experimental Framework

In order to analyze the k most probable solutions at each step of an EBNA, we deal with four test problems and take different measurements. Next, we explain in detail the components of the experiments. In [42], the necessary tools to reproduce the experiments or to carry out a similar analysis are implemented.

3.1 Problems

The whole set of problems is based on additively decomposable functions (ADFs) defined as,

$$f(\mathbf{x}) = \sum_{i=1}^{m} f_i(\mathbf{s_i}) \tag{4}$$

where $\mathbf{S_i} \subseteq \mathbf{X}$. Trying to cover a wide spectrum of applications and observe the behavior of EDAs in different scenarios, we chose these four test problems: Trap5, Gaussian Ising, $\pm J$ Ising and Max-SAT. The details of each one are introduced in the following sections. These problems are selected for several reasons. Firstly, in order to investigate the influence of multimodality in the behavior of EDAs, we deal with problems that have different number of optimal solutions. The first two problems have a unique optimum and the last two problems have several optima. Secondly, all of them are optimization problems which have been widely used to analyze EDAs [4, 19, 37]. And finally, all the problems have a different nature. Trap5 [10] is a deceptive function designed in the context of genetic algorithms [17] aimed at finding their limitations. It is a separable function and in practice can be easily optimized if the structure is known. Gaussian Ising and $\pm J$ Ising come from statistical physics domains and are instances of the Ising model proposed to analyze ferromagnetism [22]. The variables are disposed on a grid and the interactions do not allow dividing the problem into independent subproblems of bounded order efficiently [31]. The Ising problem is a challenge in optimization [19, 37] and in its general form is NP-complete [3]. Max-SAT is a variation for optimization of a classic benchmark problem in computational complexity, the propositional satisfiability or SAT. In fact, SAT was the first problem proven to be NP-complete in its general form [8]. An instance of this problem can contain a very high number of interactions among variables and, in general, it can not be efficiently divided into subproblems of bounded size in order to reach the optimum. Except for the function Trap5, we have dealt with two instances for each type of problem. We only show the results for the instance with the most relevant information.

3.2 Measurements

Our first goal is to study the relationship between the probabilistic model and the function during the search. To that end, we obtain the probabilities and function values for the k MPSs at each step of EBNA and then, we calculate the Pearson correlation coefficient ρ for each pair of samples (functions and probabilities). The correlation coefficient ρ can be expressed as,

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} \tag{5}$$

where X and Y are random variables, cov means covariance and σ is the standard deviation.

We also calculate, at each step of the algorithm, the average function values for the k MPSs and the k best individuals of the population. Moreover, we are interested in knowing the position of the optimum in the ordered set of the k MPSs at each generation. Finally, trying to observe, from a genotypic point of view, how different the k MPSs are, we calculate at each EBNA step the entropy of the k MPSs by means of adding the entropy of each variable belonging to the function,

$$H(\mathbf{X}) = -\sum_{i=1}^{n}\sum_{j=1}^{r_i} p(x_i^j)\cdot\log_2 p(x_i^j) \qquad (6)$$

where n is the number of variables and r_i is the number of states of each variable.

3.3 Parameter Configuration

Sample size is very important in order to learn Bayesian networks [16] and, hence, an important parameter in EDAs based on this type of models. Thus, we use two different population sizes in order to analyze their influence in the algorithm. Firstly, we have used the bisection method [36] to determine an adequate population size to reach the optimum (with high probability). This size is denoted by m. The stopping criterion for bisection is to obtain the optimum in 5 out of 5 independent runs. The final population size is the average over 20 successful bisection runs. The second population size is half of the bisection, $m/2$. With this size we try to create a more realistic scenario in which achieving the optimum is less likely. Thus, we can analyze the algorithm when the optimum is not reached.

The calculation of the k most probable solutions has a high computational cost. Therefore, it is necessary to limit both k and the number of function variables. In this work we deal with $k = 50$ and with two different problem sizes: dimension $n = 50$ for Trap5 and Max-SAT and dimension $n = 64$ (grid 8×8) for Gaussian Ising and $\pm J$ Ising. The stopping criterion for EBNA is a fixed number of iterations, in particular 30. It it is independent of obtaining the optimum. This number is enough to observe the convergence of the algorithm.

Finally, for each experiment type i.e. EBNA solving a problem with a given population size, 20 independent runs have been carried out. Each set of 20 executions is divided into successful (the optimum is reached) and unsuccessful (the optimum is not reached) runs which will be analyzed separately.

4 Analyzing the k MPSs in Trap5

4.1 Trap5 Description

Our first function, Trap5 [10], is an additively separable (non overlapping) function with a unique optimum. It divides the set $\mathbf{X}$ of n variables, into disjoint subsets $\mathbf{X}_I$ of 5 variables. It can be defined using a unitation function $u(\mathbf{y}) = \sum_{i=1}^{p} y_i$ where $\mathbf{y}\in\{0,1\}^p$ as,

$$\text{Trap5}(\mathbf{x}) = \sum_{I=1}^{\frac{n}{5}} trap_5(\mathbf{x}_I) \tag{7}$$

where $trap_5$ is defined as,

$$trap_5(\mathbf{x}_I) = \begin{cases} 5 & \text{if } u(\mathbf{x}_I) = 5 \\ 4 - u(\mathbf{x}_I) & \text{otherwise} \end{cases} \tag{8}$$

and $\mathbf{x}_I = (x_{5I-4}, x_{5I-3}, x_{5I-2}, x_{5I-1}, x_{5I})$ is an assignment to each trap partition $\mathbf{X}_I$. This function has one global optimum in the assignment of all ones for $\mathbf{X}$ and a large number of local optima, $2^{n/5} - 1$.

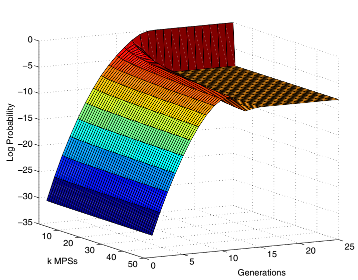
Trap5 function has been used in previous works [19] to study the structure of the probabilistic models in EDAs based on Bayesian networks as well as the influence of different parameters [25]. It is important to note that this function is difficult to optimize if the probabilistic model is not able to identify interactions between variables [12].

4.2 Experimental Results

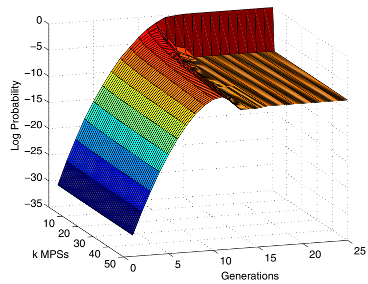
In this section, we present and discuss the results obtained when EBNA tries to optimize the Trap5 function. Firstly, in Fig. 1 and Fig. 2 we show the probabilities and function values for the k MPSs during the search and also their Pearson correlation using different population sizes respectively. We can observe that the probability distribution tends to concentrate on a unique solution at the end of the run assigning the same probability to the remainder $k-1$ MPSs. This behavior is analogous for the function values, where the MPS has the highest function value and the rest of $k-1$ MPSs have lower values. In general, the correlation rapidly increases and decreases in the first generations and grows during the rest of the search.

Particularly, a very interesting observation from these results is the difference not only between population sizes but also between successful and unsuccessful runs. Firstly, with population size $m/2$ (Fig. 2), the correlation suffers a clearer decrease in the middle of the run than with population size m. The function values for the k MPSs also reflect a difference between population sizes. Thus, with size $m/2$ (Fig. 2) the function values grow more slowly during the search. Secondly, we have different behaviors depending on the success of the search. When EBNA reaches the optimum with both population sizes, we can see that the correlation tends to 1 at the end of the run. This is independent of the number of k MPSs that we take into account. However, in unsuccessful runs the correlation at the end of the run is lower with higher values of k. Moreover, in unsuccessful runs, the correlation reaches lower values with population size $m/2$ than with m.

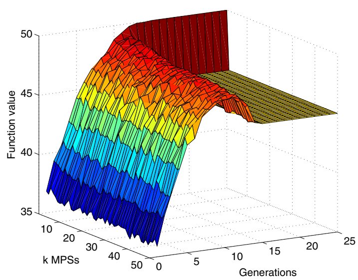
The charts of probability values in logarithmic scale have a constant pattern in all cases (Figs. 1 and 2). However, there are clear differences in the behavior of the function values between successful and unsuccessful runs in both scenarios (population sizes m and $m/2$). This difference is particularly notable in the last generations when EBNA converges to a unique solution [12]. This indicates that these function



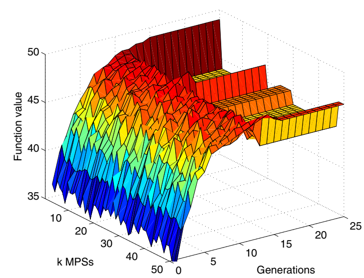
(a) Log. of the probability. Successful



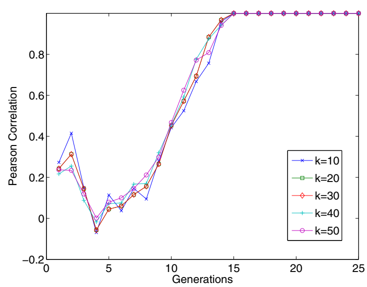
(b) Log. of the probability. Unsuccessful



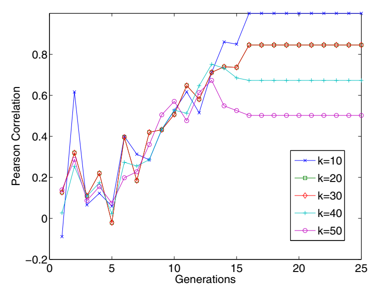
(c) Function values. Successful



(d) Function values. Unsuccessful



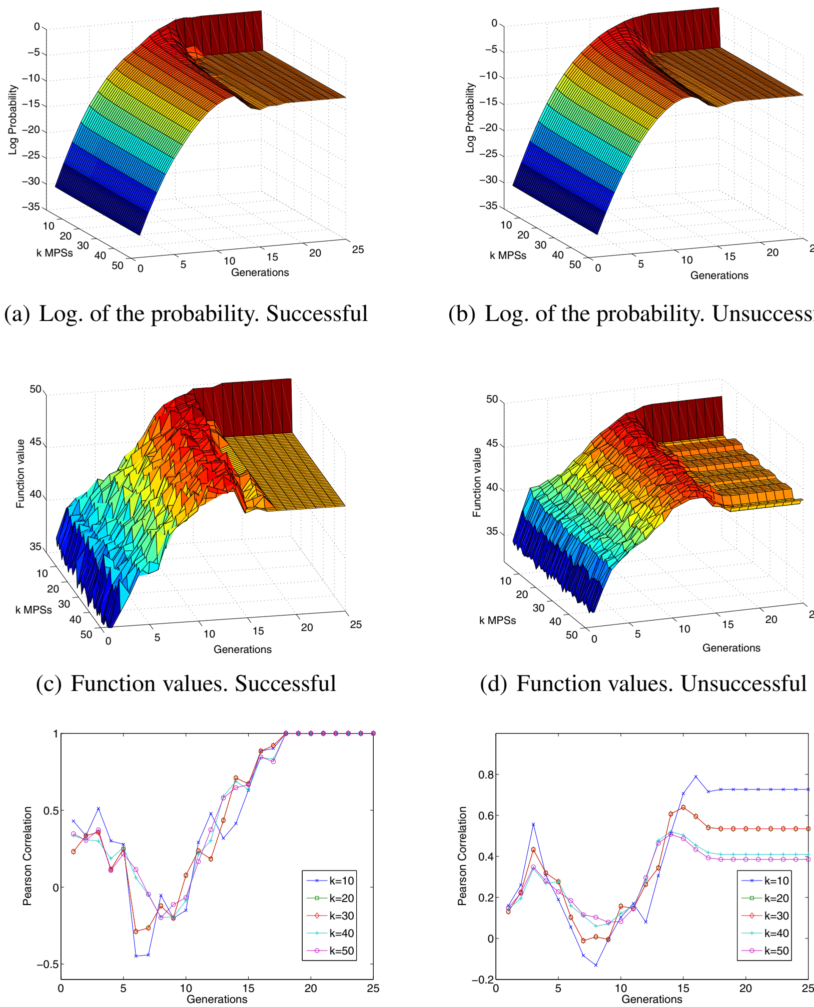
(e) Correlation. Successful



(f) Correlation. Unsuccessful

Fig. 1 Probability values, function values and Pearson correlation of the 50 most probable solutions at each generation of EBNA when it is applied to Trap5 with population size *m*. In (a), (c), (e) we provide the results for 17 successful runs out of 20. In (b), (d), (f) we provide the results for 3 unsuccessful runs

values are mainly responsible for the different behavior in the correlation between successful and unsuccessful runs. The function values for the *k* MPSs and therefore the correlation, are an important source of information to distinguish both types of runs in Trap5.



(a) Log. of the probability. Successful

(b) Log. of the probability. Unsuccessful

(c) Function values. Successful

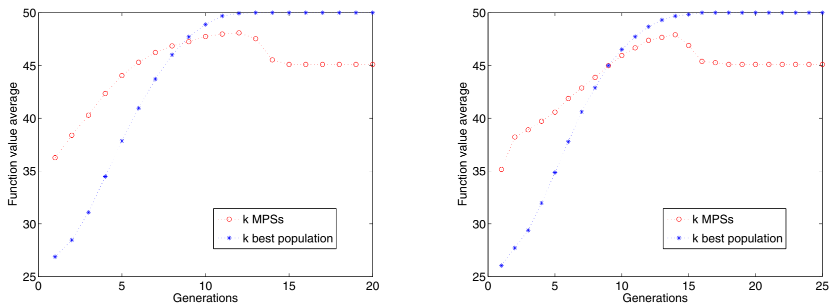
(d) Function values. Unsuccessful

(e) Correlation. Successful

(f) Correlation. Unsuccessful

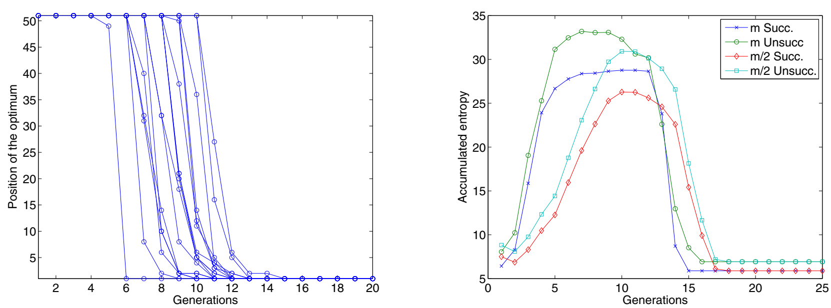
Fig. 2 Probability values, function values and Pearson correlation of the 50 most probable solutions at each generation of EBNA when it is applied to Trap5 with population size m. In (a), (c), (e) we provide the results for 3 successful runs out of 20. In (b), (d), (f) we provide the results for 17 unsuccessful runs

In Fig. 3 we report the average function values for the 50 MPSs and the 50 best individuals in the population. The results agree with [12] where the function value for the MPS and the best individual of the population are analyzed. The function values for the k MPSs are better than the k best values in the population at the beginning of the run. This difference is higher with population size m. At the end of the run, the MPSs have lower values because the EBNA population converges to a



(a) Population size m (b) Population size $m/2$

Fig. 3 Function values average for the 50 most probable solutions and the 50 best individuals of the population at each generation of EBNA when it is applied to Trap5 and the optimum is reached



(a) Position of the optimum (b) k MPSs entropies

Fig. 4 EBNA is applied to Trap5. (a) Position of the optimum in the k most probable solutions at each generation with population size m. (b) Accumulated entropy of the whole set of variables in the k most probable solutions

solution, while in the MPSs there are different solutions. For this type of analysis, the results are similar in both successful and unsuccessful runs as was shown in [12, 13].

The charts in Fig. 4 correspond to two different analysis. Firstly, Fig. 4(a) reports the position of the optimum in the ordered set of k MPSs for each run (different curves in the chart) during the search. We only show the results for successful runs because when the optimum is not reached it does not enter the k MPSs. We can see that the optimum quickly goes up in the ranking before reaching the first position. Secondly, Fig. 4(b) reports the entropy accumulated for the whole set of variables in the k MPSs at each generation. We can observe a constant pattern in the four cases reported. The entropy increases at the beginning of the search and decreases in the last generations when EBNA converges to a solution. In the first and last generations, when the Bayesian networks have the lowest complexity, the k MPSs have very similar genotypes. Nevertheless, in the middle of the run the entropy increases with the complexity of the Bayesian network. It is also important to note that it is possible to distinguish between successful and unsuccessful runs in these results. The k MPSs are more entropic in the middle of the run when the optimum is not reached. This occurs for both population sizes m and $m/2$.

5 Analyzing the k MPSs in Gaussian Ising

5.1 2D Ising Spin Glass Description

Ising spin glass model [22] is often used as optimization problem in benchmarking EDAs [19, 37, 39]. A classic 2D Ising spin glass can be formulated in a simple way. The set of variables $\mathbf{X}$ is seen as a set of n spins disposed on a regular 2D grid L with $n = l \times l$ sites and periodic boundaries (see Fig. 5). Each node of L corresponds to a spin X_i and each edge (i, j) corresponds to a coupling between X_i and X_j. Thus, each spin variable interacts with its four nearest neighbors in the toroidal structure L. Moreover, each edge of L has an associated coupling strength J_{ij} between the related spins. For the classical Ising model each spin takes the value 1 or -1. The target is, given couplings J_{ij}, to find the spin configuration that minimizes the energy of the system computed as,

$$E(\mathbf{x}) = -\sum_{(i,j)\in L} x_i J_{ij} x_j - \sum_{i\in L} h_i x_i \tag{9}$$

where the sum runs over all coupled spins. In our experiments we take $h_i = 0 \; \forall i \in L$. The states with minimum energy are called *ground states*.

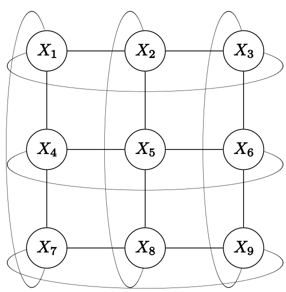


Fig. 5 A 3×3 grid structure L showing the interactions between spins for a 2D Ising spin glass with periodic boundaries. Each edge has an associated strength J_{ij}

Depending on the range chosen for the couplings J_{ij} we have different versions of the problem. Thus, the problem is called *Gaussian Ising* when the couplings J_{ij} are real numbers generated following a Gaussian distribution. For this type of couplings, the problem has only one optimal solution. A specified set J_{ij} of coupling defines a spin glass instance. We generated the Gaussian Ising instances using the Spin Glass Ground State server[1] for the experiments. The minimum energy of the system is also provided in this server.

5.2 Experimental Results

In this section, we present and discuss the results obtained when EBNA tries to solve the Gaussian Ising problem. In Fig. 6, we report the probabilities, function values and correlations for the k MPSs with the population size given by bisection. In Gaussian Ising, the results for population size $m/2$ are not shown because they do not provide new relevant information. In contrast with Trap5, the first type of analysis for this problem does not show a clear difference between either population sizes or between successful and unsuccessful runs. Nonetheless, for this last matter we can observe little differences in the correlations (Fig. 6(c) and 6(f)) where the curves for all k reach very close values in unsuccessful runs. Moreover, in the charts of function values (Fig. 6(b) and 6(e)), when EBNA does not reach the optimum, the MPSs with the lowest probability (they are always ordered from 1 to 50) have worse function values in the last generations.

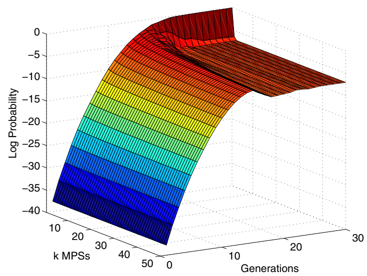
In any case, the different behavior of the k MPSs between these two problems is evident. Firstly, in Gaussian Ising, although the probability distribution has an analogous behavior, it concentrates on a unique solution more slowly than in Trap5 in the last generations. Secondly, there is a clear difference in the behavior of the function values for the k MPSs. In Trap5, the MPS always has a better function value than the rest of $k-1$ solutions at the end of the run. In Gaussian Ising, however, there is a clear diversity in the function values for the k MPSs and their form is barely related with the probability values. Lastly, these facts are reflected in the charts of correlation. In Gaussian Ising, the curve of the correlation has a lower slope and reaches lower values than in Trap5 at the end of the run.

Although both problems have a unique optimum, we believe that the difference in the behavior of the k MPSs is due to the properties of the landscape for each problem. In Gaussian Ising few solutions share the same function value, while in Trap5 there are different sets of solutions with the same function value.

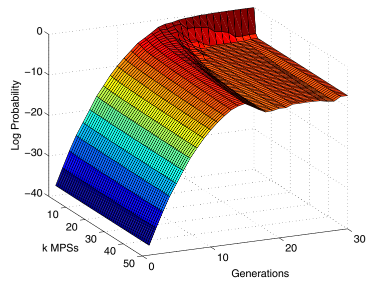
In Fig. 7 we report the average function value for the 50 MPSs and the 50 best individuals in the population. Once again, the results agree with [12]. In this problem, there is a shorter distance than in Trap5 between both curves during the search. Moreover, there is a lower difference between both population sizes. Nevertheless, the average function value for the k MPSs is better than for the k best individuals in the population during the first generations.

As in the previous section, Fig. 8(a) reports the position of the optimum in the k MPSs during the search. In general, this analysis shows a behavior similar to Trap5. However, in this case, the optimum can decrease in position from one generation to the next. This can be seen because some curves have upward lines. Regarding the entropies of the k MPSs, we present Fig. 8(b). In this problem the behavior of the

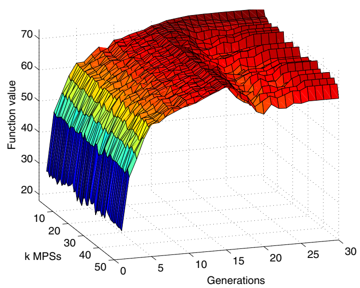
[1] `http://www.informatik.uni-koeln.de/ls_juenger/index.html`



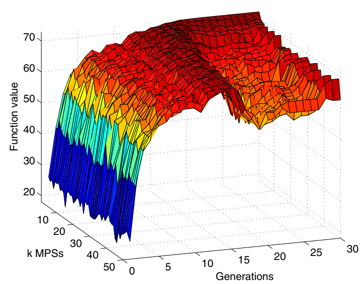
(a) Log. of the probability. Successful



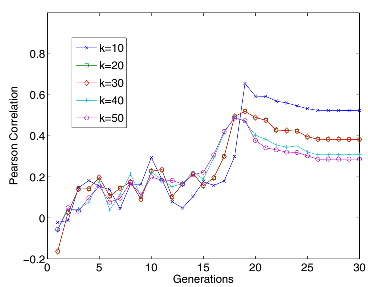
(b) Log. of the probability. Unsuccessful



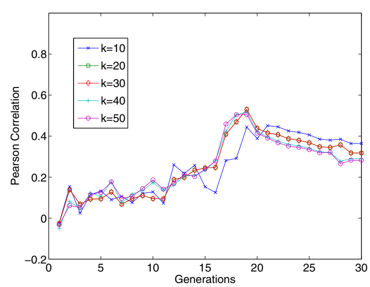
(c) Function values. Successful



(d) Function values. Unsuccessful



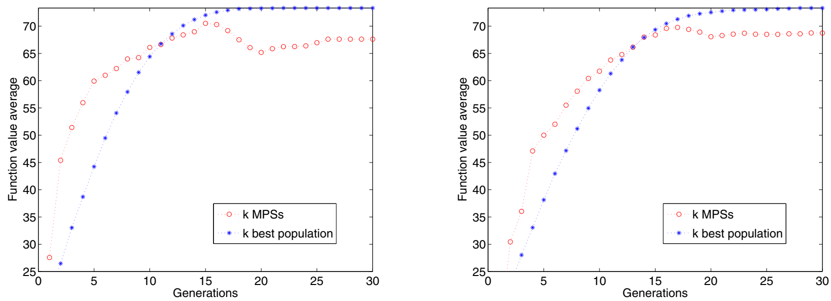
(e) Correlation. Successful



(f) Correlation. Unsuccessful

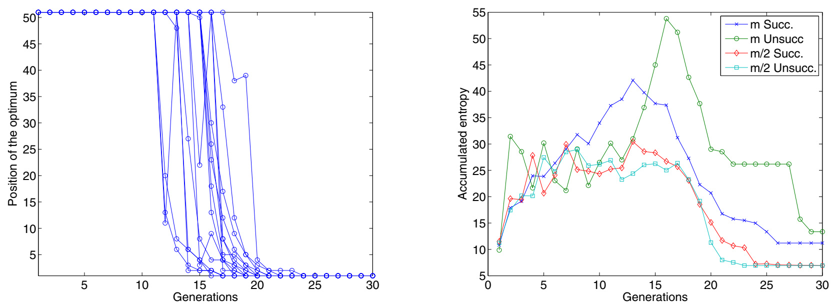
Fig. 6 Probability values, function values and Pearson correlation of the 50 most probable solutions at each generation of EBNA when it is applied to Gaussian Ising with population size m. In (a), (c), (e) we provide the results for 15 successful runs out of 20. In (b), (d), (f) we provide the results for 5 unsuccessful runs

different curves is more variable and the pattern in general is less clear than in Trap5. Nevertheless, this chart allows us to distinguish between different scenarios. First of all, with population size m we can distinguish between successful and unsuccessful runs. This is because the entropy of the k MPSs in unsuccessful runs reaches higher values. This behavior was also observed for Trap5. Secondly, we can observe a different behavior with population size m and $m/2$. The curves for m clearly increase



(a) Population size m (b) Population size $m/2$

Fig. 7 Function values average for the 50 most probable solutions and the 50 best individuals of the population at each generation of EBNA when it is applied to Gaussian Ising and the optimum is reached



(a) Position of the optimum (b) k MPSs entropies

Fig. 8 EBNA is applied to Gaussian Ising. (a) Position of the optimum in the k most probable solutions at each generation with population size m. (b) Accumulated entropies of the whole set of variables in the k most probable solutions

from the beginning of the run to the middle and decrease in the last generations, while for population size $m/2$ the curves have a lower growth.

6 Analyzing k MPSs in $\pm J$ Ising

6.1 $\pm J$ Ising Description

As explained in Section 5, the main difference between both versions of $2D$ Ising spin glass is the range of values chosen for the couplings J_{ij}. In this second Ising problem, the couplings J_{ij} are set randomly to either $+1$ or -1 with equal probability. This version, that will be called $\pm J$ *Ising*, could have different configurations of the spins that reach the ground state (lowest energy) and therefore many optimal solutions may arise. As for the previous case, the $\pm J$ Ising instances were generated using the Spin Glass Ground State server. This server also provided the value of the minimum energy of the system.

6.2 Experimental Results

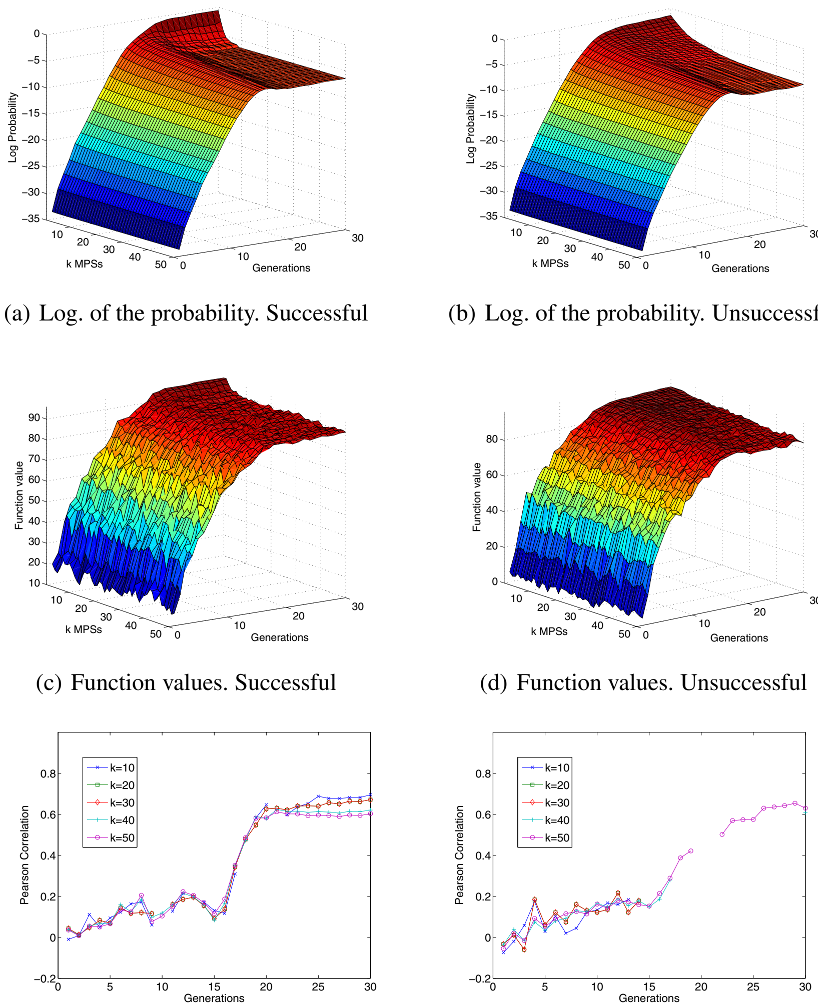
In this section, we present and discuss the results obtained when EBNA tries to solve the $\pm J$ Ising problem. The analyisis of the probabilities, function values and correlation for the k MPSs are shown in Fig. 9. In this problem the results with different population sizes are very similar. We only report those with population size $m/2$ in order to have a wider variety of successful and unsuccessful runs. As we previously explained, $\pm J$ Ising problem has several optimal solutions and this fact is reflected in these results.

First for all, we must explain a particular behavior which can arise in the problems with several optima. If we look at Fig. 9(f), we can see that there is only one curve ($k = 50$) in the last generations and it is not continuous. The reason for such behavior is that in certain generations of some runs, all the function values for the k MPSs are equal. In this case, the standard deviation is 0 and therefore Equation 5 makes no sense. This situation will be more evident in the Max-SAT problem which is analyzed in the following section. Although we do not directly analyze this matter, it occurs more frequently in unsuccessful runs. This can also be intuited through the charts of function values. Thus, if we compare Figs. 9(b) and 9(e), we can see that in unsuccessful runs the function values have very little diversity in the last generations. However, for successful runs we can appreciate different function values from the first k MPSs to the last. This is reflected in the correlation charts where all curves are complete for successful runs. In this case, we have a low correlation during the search and it just increases in the last generations when the algorithm converges.

In this problem with several optima, we can observe, for the first time, differences between successful and unsuccesful runs by looking at the probability values of the k MPSs. Depending on the type of run, the probability is distributed in a different manner in the last generations. Thus, in successful runs the probability is more concentrated in the first MPSs.

For this problem, Fig. 10 shows a similar behavior than for Gaussian Ising. Once again the function values for the k MPSs outperform the k best individuals in the population at the beginning of the run. We can also see a difference between population sizes. The curves have a greater distance in the first generations with a population size m.

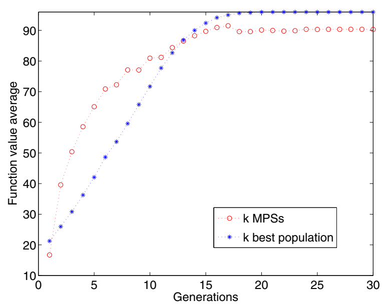
Finally, in Fig. 11 we report the position of the optimum in the k MPSs and their entropy during the search. In Fig. 11(a), where EBNA solves $\pm J$ Ising with population size m, the behavior is similar to that in Gaussian Ising. The curves have similar slopes and the optimum can decrease in the ranking from one generation to the next. We have selected this population size in order to have more successful runs (more curves) and therefore more variety. In Fig. 11(b), the entropy curves are also similar to those for Gaussian Ising. We can only distinguish between successful and



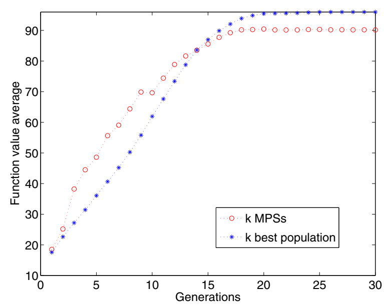
(a) Log. of the probability. Successful
(b) Log. of the probability. Unsuccessful
(c) Function values. Successful
(d) Function values. Unsuccessful
(e) Correlation. Successful
(f) Correlation. Unsuccessful

Fig. 9 Probability values, function values and Pearson correlation of the 50 most probable solutions at each generation of EBNA when it is applied to $\pm J$ Ising with population size $m/2$. In (a), (c), (e) we provide the results for 8 successful runs out of 20. In (b), (d), (f) we provide the results for 12 unsuccessful runs

unsuccessful runs when the population size given by bisection is used. In this case, the k MPSs are clearly more entropic in unsuccessful runs at the end of the run. Regarding the population size, we can see that with size m, there is more genotypic diversity during the search.

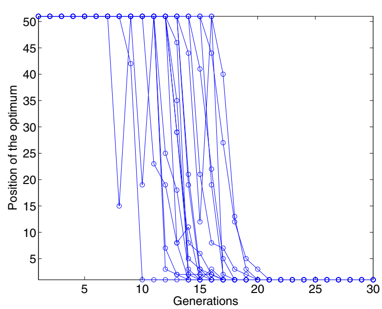


(a) Population size m

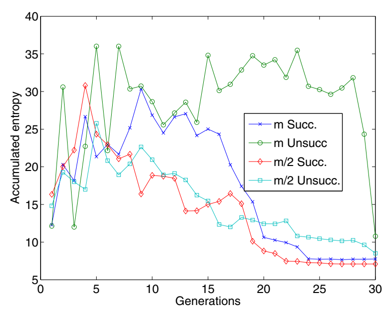


(b) Population size $m/2$

Fig. 10 Function values average for the 50 most probable solutions and the 50 best individuals of the population at each generation of EBNA when it is applied to $\pm J$ Ising and the optimum is reached



(a) Position of the optimum



(b) k MPSs entropies

Fig. 11 EBNA is applied to $\pm J$ Ising. (a) Position of the optimum in the k most probable solutions at each generation with population size m. (b) Accumulated entropies of the whole set of variables in the k most probable solutions

7 Analyzing k MPSs in Max-SAT

7.1 Max-SAT Description

The last problem in our analysis is the maximum satisfiability or Max-SAT problem, which has often been used in different works about EDAs [4, 37]. Without going into details, given a set of Boolean variables $\mathbf{X}$ and a Boolean expression ϕ, SAT problem asks if there is an assignment $\mathbf{x}$ of the variables such that the expression ϕ is satisfied. In a Boolean expression we can combine the variables using Boolean connectives such as $\wedge$ (logical and), $\vee$ (logical or) and $\neg$ (negation). An expression of the form x_i or $\neg x_i$ is called a literal.

Every Boolean expression can be rewritten into an equivalent expression in a convenient specialized style. In particular, we use the *conjunctive normal form* (CNF) $\phi = \bigwedge_{i=1}^{q} C_i$. Each of the q C_is is the disjunction of two or more literals which are called clauses of the expression ϕ. We work with clauses of length $k = 3$. When $k \geq 3$, the SAT problem becomes NP-Complete [8]. An example of a CNF expression with 5 Boolean variables X_1, X_2, X_3, X_4, X_5 and 3 clauses could be, $\phi = (x_1 \vee \neg x_3 \vee x_5) \wedge (\neg x_1 \vee x_3 \vee x_4) \wedge (x_1 \vee \neg x_4 \vee \neg x_2)$.

The Max-SAT problem has the same structure as SAT, but the result, for an assignment $\mathbf{x}$, is the number of satisfied clauses instead of a truth value. In order to solve Max-SAT, the assignment for $\mathbf{X}$ that maximizes the number of satisfied clauses must be found. Thus, the function can be written as,

$$f_{Max-SAT}(\mathbf{x}) = \sum_{i=1}^{q} \phi(C_i) \tag{10}$$

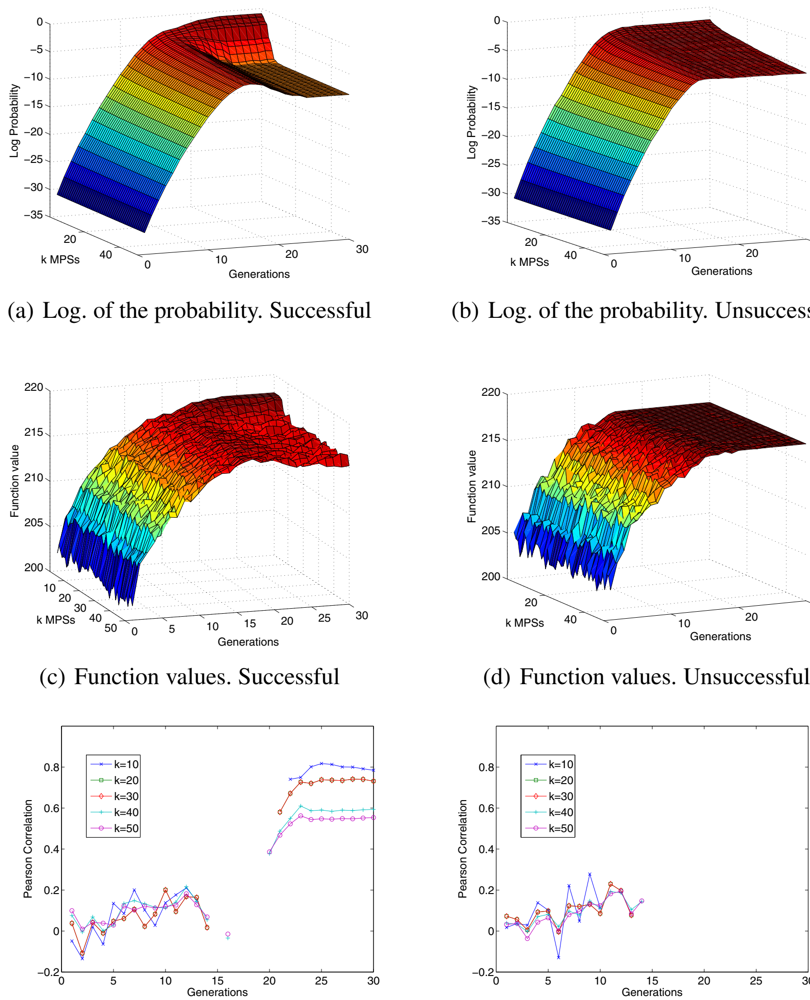
where each clause C_i of three literals is evaluated as a Boolean expression that returns 1 if the expression is *true* and 0 if it is *false*. Since C_i is a disjunction, it is satisfied if at least one of its literals is *true*. The variables of $\mathbf{X}$ can overlap arbitrarily in the clauses.

Particularly, we work with 3-CNF SAT problems obtained from the SATLIB [21] repository which provides a large number of SAT instances. The instances used are satisfiable. They have 50 variables and 218 clauses. It is important to note that there could be several assignments for $\mathbf{X}$ that satisfy all clauses and therefore this problem could have different optimal solutions.

7.2 Experimental Results

In this section, we present and discuss the results obtained when EBNA tries to solve the Max-SAT problem. As with $\pm J$ Ising, this problem has several optimal solutions and it is reflected in the analysis. In Fig. 12 we present the probabilities in logarithmic scale, the function values and their correlation when population size $m/2$ is used. Max-SAT has a similar behavior for both population sizes in this type of analysis. So only results for the most representative size ($m/2$) are shown.

We can find clear differences between successful and unsuccessful runs in all the charts of Fig. 12. Firstly, regarding the probability values, at the end of the run it is evident that the probability is concentrated on the first MPSs when EBNA reaches the optimum. By contrast, the probability is distributed more equitably among the k MPSs in unsuccessful runs. Secondly, if we look at the charts of function values, we can see that almost all k MPSs have the same value in the last generations for unsuccessful runs. On the contrary, when the optimum is reached, there is diversity in the function values for this set of solutions. Lastly, in the charts of correlation we have the same problem as in $\pm J$ Ising because in some generations the calculation of the correlation is not possible. This problem is more evident in unsuccessful runs where we can see that the function values reach equal values in most of the cases. Once again, there is a low correlation between probabilities and functions during the



(a) Log. of the probability. Successful

(b) Log. of the probability. Unsuccessful

(c) Function values. Successful

(d) Function values. Unsuccessful

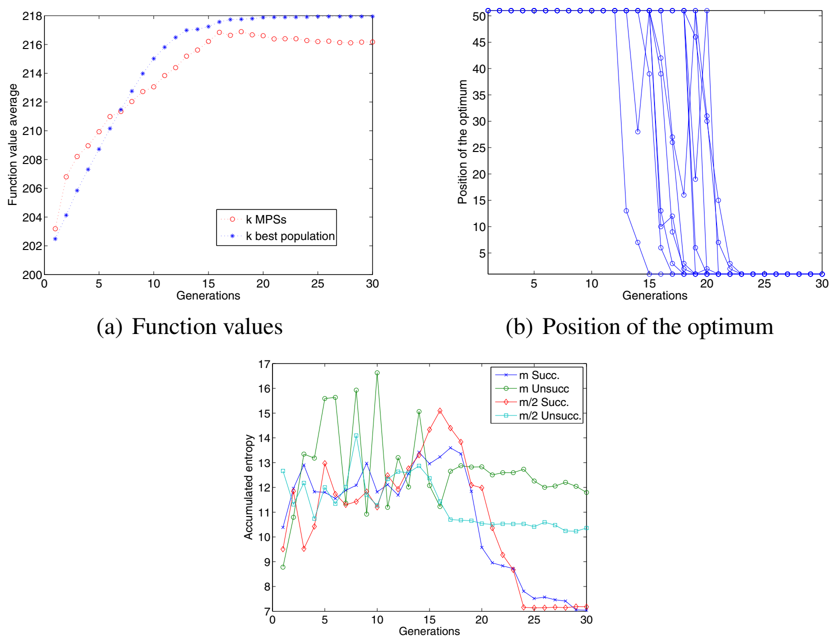
(e) Correlation. Successful

(f) Correlation. Unsuccessful

Fig. 12 Probability values, function values and Pearson correlation of the 50 most probable solutions at each generation of EBNA when it is applied to Max-SAT with population size $m/2$. In (a), (c), (e) we provide the results for 12 successful runs out of 20. In (b), (d), (f) we provide the results for 8 unsuccessful runs

search. Only at the end of the run, when the optimum is reached, does the correlation increase (see Fig. 12(c)).

In Fig. 13 we show the rest of the analysis. A little difference between population sizes in the analysis of the average function value can be seen. Therefore, we only show Fig. 13(a) as an example of the behavior. We can see that the k MPSs are only better than the k best individuals of the population in few generations at the beginning of the run. In the rest of the run there are better individuals in the population. It is possible that Bayesian networks do not manage to accurately represent the structure of the problem. Fig. 13 shows the position of the optimum in the k MPSs during the search and the behavior is the same as in the rest of the problems. Finally, the entropies of the k MPSs in Fig. 13(c) reveal a clear difference between successful and unsuccessful runs. Thus, when the optimum is not reached by EBNA our analyzed solutions are more entropic at the end of the run. However, in this chart it is difficult to distinguish between population sizes.



(a) Function values

(b) Position of the optimum

(c) k MPSs entropies

Fig. 13 EBNA is applied to Max-SAT. (a) Function values average for the 50 MPSs and the 50 best individuals of the population for successful runs with population size m. (b) Position of the optimum into the 50 MPSs at each generation with population size $m/2$. (c) Accumulated entropy of the whole set of variables in the 50 MPSs

8 Related Works

Most of the research done in the scope of the models learned by EDAs that use Bayesian networks [15, 30, 38] has focused on structural descriptors of the networks, in particular on the type (i.e. correct or spurious) and number of the network edges [11, 14, 18, 19, 24, 25]. The analysis of the Bayesian network edges learned by EDAs has allowed to study the effect of the selection and replacement [19, 25] as well as the learning method [11, 14, 24] in the accuracy of the models learned by EDAs and the efficiency of these algorithms. A more recent work [24] considers the likelihood given to the selected set during the model learning step as another source of information about the algorithm's behavior.

We are in the line of works which analyze EDAs from a quantitative point of view. For EDAs that use Markov models, in [5] the product moment correlation coefficient between the Markov model learned by DEUM and the fitness function is used to measure the quality of the model as a fitness function predictor. For a given solution, the prediction is the value given by the Markov model to the solution. The quality of the model is measured using the correlation computed from samples of the search space.

For EDAs based on Bayesian networks, in [12, 13] the probability of the optimum is analyzed. By using the most probable solution given by the probabilistic model, the optimum of the function and the best individual of the population, basic issues about the behavior of EDAs are studied. In this chapter, we extend this type of quantitative analysis, taking into consideration the k MPSs as descriptors of the probabilistic models. The computation of the MPS and the k MPSs has also been used as a way to improve the sampling step in EDAs [20, 28]. In this context, different questions arise such as which is a good value for k, at which time of the evolution it is more convenient to introduce the k best MPSs, how to avoid premature convergence, etc. The kind of analysis we present may be useful to answer these questions and advance in this line of research.

9 Conclusions

In this work, we have exploited the quantitative component of the probabilistic models learned by EDAs during the search in order to better understand their behavior and aid in their development. Particularly, we have analyzed the k solutions with the highest probability in the distributions estimated at each step of an EBNA. We have conducted systematic experiments for several functions.

Firstly, we have studied the relation between the probabilistic model and the function. For the k MPSs, we have recorded the probabilities, the function values and the correlation between both at each step of the algorithm. We have observed that the results change markedly depending on the problem. This allows us to discover some properties of the search space. Firstly, it is possible to distinguish multimodal functions such as $\pm J$ Ising and Max-SAT. The main source to identify this characteristic in the function are the probability values. Thus, when the function has a unique optimum, the probability is concentrated in the MPS at the end of the run. However, when the function is multimodal, the probability is shared among several k MPSs in the last generations. This supports the conclusions drawn in [13]. Secondly, the function values for the k MPSs are a rich source of information in the problems with a unique optimum (Trap5 and Gaussian Ising). Although the probability values have a similar behavior in both problems, the function values provide information about the properties of the search space. Thus, we have seen that Trap5 has many local optima with the same function value. However, in Gaussian Ising, the function values for the k MPSs suffer important variations even at the end of the run. Therefore, we can deduce that this function assigns more different function values than Trap5 to the search space. The correlation between probabilities and function values has a different curve depending on the problem. Only in Trap5 does it reach the value of 1 in the last generations.

It is very important to note that this analysis reveals a different behavior between successful and unsuccessful runs in most of the cases. For this goal, the function values and the correlations are the main source of information in the problems with a unique optimum. When the optimum is not reached, the function values suffer variations and the correlation is lower at the end of the run. Although in Gaussian Ising there is only a little difference between both types of runs, in Trap5 the successful runs are easily identifiable. In problems with several optima, both the probability and function values, and hence the correlations, allow us to identify the success of the search. Firstly, the probability values are more spread among the *k* MPSs when the optimum is not reached. This fact is more evident in Max-SAT. Secondly, the function values for these solutions are more variable in successful runs. By contrast, when the optimum is not reached, it is more frequent that all *k* MPSs have equal function values in a given generation. Therefore, this affects the calculation of the correlation. Thus, in unsuccessful runs, there are more generations for which the correlation has no solution. The distinction between types of runs is a very important matter. It could open a line of work in order to create methods to predict or estimate the success of a particular run in real problems where the optimum is not known.

Regarding the analysis of the average function value for the *k* MPSs and the *k* best individuals of the population, we have observed that they agree with the results obtained in [12]. Thus, the function values for the *k* MPSs always have a higher quality than the best individuals of the population at the beginning of the run. In general, this difference is greater when a population size given by bisection is used in EBNA. However, in the middle and at the end of the run the behavior depends on the problem. In general, the behavior observed in this type of analysis confirms the benefit of exploiting the information that the Bayesian network contains about the function by using inference techniques [28], at least at the beginning of the run.

The position of the optimum inside the *k* MPSs during the search has a similar behavior in different problems. Thus, the optimum usually reaches the first position in few generations when it is reached. Except for Trap5, it is possible for the optimum to go down in the ranking from one generation to the next. Moreover, although these results have not been presented, it is also possible for the optimum to enter the *k* MPSs and to leave in the next generation. Once again, this supports the idea of using approximate inference techniques of the MPSs in the sampling step of an EDA.

Finally, the analysis of the entropies for the *k* MPSs allows us to make important distinctions not only between problems but also between successful and unsuccessful runs and population sizes. A common pattern in all problems is that the entropy of the *k* MPSs is lower in successful runs, especially in the middle of the run. It indicates that the probability distribution assigns the highest probabilities to more homogeneous solutions and it is beneficial for the search. Regarding the difference between population sizes, the entropy tends to be lower with a reduced size.

We can assert that the k MPSs are a rich source of information about the behavior of an EDA. However, their computation requires a high computational cost. We believe that carrying out an approximate calculation of the k MPSs could be useful to obtain relevant information about the optimization process. This type of measurements could be taken on-line during the search allowing certain automatic decision making in the algorithm which could be really useful to develop adaptive EDAs.

It is important to highlight that our method may be extended to EDAs that use other classes of probabilistic models by modifying the class of message passing algorithm used in the computation of the MPSs (e.g. using loopy belief propagation). Furthermore, it is also possible to use information about the fitness function (structure and fitness values) [20, 26, 41] to conveniently modify the definition of the abductive inference step and the computation of the MPSs.

Acknowledgments

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